

ORF307: Optimization

Spring 2023 1st Midterm Exam Solutions

March 9, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (40 points).* Multiobjective least squares with an affine term. Given a set of k vectors $a_1, \dots, a_k \in \mathbf{R}^n$, consider a version of multiobjective least squares that adds an affine term $c^T x + d$ for $c \in \mathbf{R}^n$, $d \in \mathbf{R}$. Specifically, consider the optimization problem:

$$\text{minimize } c^T x + d + \sum_{i=1}^k \|x - a_i\|^2. \quad (1)$$

- (a) (15 points) Define A and b so that we can equivalently express problem (1) as

$$\text{minimize } \|Ax - b\|^2 + c^T x + d, \quad (2)$$

and also give the dimensions of A and b .

- (b) (15 points) Show that problem (2) can be rewritten as

$$\text{minimize } \|Ax - b + f\|^2 + g, \quad (3)$$

for some f and g . *Hint: Expand the squared 2-norms and recall that $u^T v = v^T u$ for any vectors u, v . Also, f and g may not be unique. Specifically, in your derivation, argue that the two objectives can be made equal if $A^T f = c/2$. Then, either justify why f must exist or pick a valid choice of f .*

- (c) (10 points) Now suppose we add constraints $Cx = h$ to find a new problem:

$$\begin{aligned} &\text{minimize } \|Ax - b + f\|^2 + g, \\ &\text{subject to } Cx = h. \end{aligned} \quad (4)$$

Write the KKT linear system for problem (4). Your final answer should be in terms of $\{a_i\}_{i=1}^k, k, c, C, h$, and the solution vector.

Solution.

- (a) As done in class, recognize that we can stack the squared 2-norms:

$$\sum_{i=1}^k \|x - a_i\|^2 = \left\| \begin{bmatrix} x - a_1 \\ \vdots \\ x - a_k \end{bmatrix} \right\|^2 = \left\| \begin{bmatrix} I_n \\ \vdots \\ I_n \end{bmatrix} x - \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} \right\|^2.$$

So, we choose

$$A = \begin{bmatrix} I_n \\ \vdots \\ I_n \end{bmatrix}, \quad b = \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix}.$$

Specifically, I_n is the n -dimensional identity matrix, and A is the stack of k copies of I_n while b is the stack of the a_i 's. So, $A \in \mathbf{R}^{kn \times n}$ and $b \in \mathbf{R}^{kn}$.

(b) We expand both objectives

$$\begin{aligned}\|Ax - b\|^2 + c^T x + d &= (Ax - b)^T (Ax - b) + c^T x + d \\ &= x^T A^T A x - (2b^T A - c^T)x + b^T b + d,\end{aligned}$$

and

$$\begin{aligned}\|Ax - b + f\|^2 + g &= (Ax - b + f)^T (Ax - b + f) + g \\ &= x^T A^T A x - 2(b - f)^T A x + (b - f)^T (b - f) + g.\end{aligned}$$

By matching terms, we have:

$$\begin{aligned}2b^T A - c^T &= 2b^T A - 2f^T A, \\ b^T b + d &= (b - f)^T (b - f) + g.\end{aligned}$$

The first equation tells us that $A^T f = c/2$. Since A is tall, A^T is wide, so we have infinite choices for f (for $k > 1$). One such valid choice is $f = (c/(2k), \dots, c/(2k))$. Once we fix f , we can also find the corresponding g value since

$$g = b^T b + d - (b - f)^T (b - f).$$

(c) We know

$$A^T A = \begin{bmatrix} I_n & \dots & I_n \end{bmatrix} \begin{bmatrix} I_n \\ \vdots \\ I_n \end{bmatrix} = \sum_{i=1}^k I_n = kI_n,$$

and

$$A^T b = \begin{bmatrix} I_n & \dots & I_n \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} = \sum_{i=1}^k a_i.$$

Lastly, even though f may have infinite choices, we know from our derivation in part (b) that $A^T f = c/2$. So, we have that

$$A^T (b - f) = -c/2 + \sum_{i=1}^k a_i.$$

Plugging all of this in and applying the KKT conditions gives the desired result:

$$\begin{bmatrix} 2kI_n & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x^* \\ z^* \end{bmatrix} = \begin{bmatrix} -c + 2 \sum_{i=1}^k a_i \\ h \end{bmatrix}.$$

2. *Problem 2 (30 points).* Difference of squared distances. A weighted difference of squared distances from x to two fixed vectors c and d is defined

$$f(x) = \|x - c\|^2 - \lambda \|x - d\|^2.$$

- (a) (2 points) Is it a multi-objective least squares problem? Explain your answer.
- (b) (6 points) Give the value(s) of λ for which $f(x)$ is affine.
- (c) (4 points) Set λ to a value found in part (b). Give the affine representation of $f(x)$, i.e., a and b for which $f(x) = a^T x + b$ holds for all x .
- (d) (6 points) We now want to minimize $f(x)^2$ ($f(x)$ from part (c)) subject to linear constraints $Px = h$. Write this problem as a constrained least squares problem of the form

$$\begin{aligned} &\text{minimize} && \|\tilde{A}x - \tilde{b}\|^2 \\ &\text{subject to} && \tilde{C}x = \tilde{d}. \end{aligned}$$

- (e) (6 points) State the KKT equations for this problem in terms of the components of $f(x)$ and of the linear constraints.
- (f) (6 points) Let the dimension of x be n , and the dimension of P be $p \times n$. What are the conditions for which the KKT matrix is invertible? You should refer to the dimensions in your answer.

Solution.

- (a) No because the second term is multiplied by $-\lambda$, which can be negative. In multi-objective least squares you sum the components with *positive* weights.
- (b) Expand the squares.

$$\begin{aligned} f(x) &= (x - c)^T(x - c) - \lambda(x - d)^T(x - d) \\ &= (x^T x - c^T x - x^T c + c^T c) - \lambda(x^T x - d^T x - x^T d + d^T d) \\ &= x^T x - 2c^T x + c^T c - \lambda(x^T x - 2d^T x + d^T d) \quad \lambda = 1 \text{ to cancel out the squares} \\ &= x^T x - 2c^T x + c^T c - x^T x + 2d^T x - d^T d \\ &= 2(d - c)^T x + \|c\|^2 - \|d\|^2 \end{aligned}$$

We must have $\lambda = 1$.

- (c) $a = 2(d - c)$, $b = \|c\|^2 - \|d\|^2$.
- (d) The problem can be written as

$$\begin{aligned} &\text{minimize} && \|a^T x - (-b)\|^2 \\ &\text{subject to} && Px = h. \end{aligned}$$

(e) The KKT equations are

$$\begin{bmatrix} 2aa^T & P^T \\ P & 0 \end{bmatrix} \begin{bmatrix} x^* \\ z \end{bmatrix} = \begin{bmatrix} -2ab \\ h \end{bmatrix}.$$

(f) For the KKT matrix to be invertible, P must have linearly independent rows. In addition, $n \geq p \geq n - 1$.

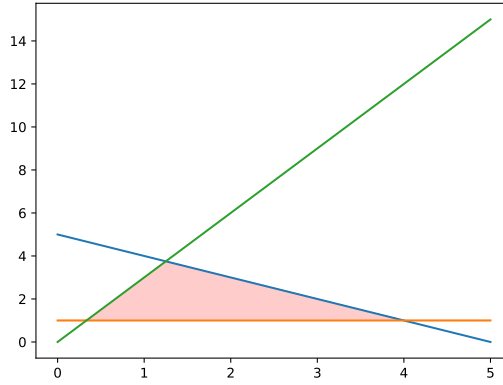
3. *Problem 3 (30 points).* Chocolate shopping. You are going shopping for chocolate with your friends Derek and Emma. The chocolates being sold by the chocolate store are Cadbury's and Hersheys. One bar of Carbury's costs \$3 while one bar of Hersheys costs \$1. Emma cares about your health so tells you that you may buy no more than 5 bars of chocolate in total. You are a fan of Hersheys, so you insist that you must purchase at least one Hersheys bar. Derek understands your love of Hersheys but is a Cadbury's fan, so he insists that the amount of Hersheys you buy is at most 3 times the amount of Cadbury's. You are shopping at an *elite Chocolatier* so you are able to purchase fractional bars of chocolate. Since you are a student of Operations Research and Financial Engineering, your objective is to purchase a quantity of each chocolate which satisfies yours, Derek's and Emma's constraints while minimizing total expenditure.
- (a) (12 points) Write the problem of purchasing the optimal amount of each chocolate as an optimization problem.
 - (b) (12 points) Draw the feasible set, draw contour lines for the objective, and use your drawing to find the optimal amount of each chocolate which you should buy.
 - (c) (6 points) Write the optimization problem which you have just formulated in standard form.

Solution.

- (a) Let x_1 be the amount of Cadbury's which you purchase, and x_2 be the amount of Hersheys. The optimization problem can then be written in the following way.

$$\begin{aligned}
 &\text{minimize} && 3x_1 + x_2 \\
 &\text{subject to} && x_1 + x_2 \leq 5 \\
 &&& 3x_1 \geq x_2 \\
 &&& x_2 \geq 1 \\
 &&& x_1 \geq 0 \\
 &&& x_2 \geq 0.
 \end{aligned}$$

- (b) The feasible region is the following.



Since all terms in the objective are positive, the optimal solution is clearly given by the point in the bottom-left corner. That is, $x^* = (1/3, 1)$.

- (c) Note that the $x_1 \geq 0$ and $x_2 \geq 0$ constraints are redundant. The feasible region is equivalent to $Ax \leq b$ where,

$$A = \begin{bmatrix} 1 & 1 \\ -3 & 1 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 0 \\ -1 \end{bmatrix}.$$

Introduce new variables s_1, s_2 and s_3 and define,

$$\tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} A & I_3 \end{bmatrix}, \quad \tilde{b} = b, \quad \tilde{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Then the equivalent standard-form problem to our original problem is,

$$\begin{aligned} & \text{minimize} && \tilde{c}^T \tilde{x} \\ & \text{subject to} && \tilde{A} \tilde{x} = \tilde{b} \\ & && \tilde{x} \geq 0. \end{aligned}$$